Case study - data platform for Eurasia Retail Group

Customer and challenge

Eurasia Retail Group is a fictional retail chain with 310 stores in Kazakhstan and Uzbekistan. In 2024 the customer needed a single data platform for sales analytics and demand forecasting, replacing spreadsheets prepared manually by regional teams. The project ran for 10 months, from January to October 2024.

Solution

Sales events from cash registers are streamed through Apache Kafka into a data lake, processed with Apache Spark and loaded into ClickHouse for reporting. The platform processes about 2 TB of data per day. Forecasting models are written in Python and managed with MLflow. Dashboards are built in Apache Superset.

Team

The team had 12 people: 4 data engineers, 3 machine learning engineers, 2 analysts, 1 DevOps engineer, 1 QA engineer and 1 project manager.

Results

The weighted mean absolute percentage error (WMAPE) of weekly demand forecasts decreased from 21% to 13%. Preparation of the weekly sales report was reduced from 2 working days to 15 minutes. Out-of-stock situations in the pilot group of 40 stores decreased by 17%.